PHYTOFUSION AI – SPEAKER 1

From Pharmacy to Plant Genetics

1. Personal introduction and opening

Good morning everyone!

Before introducing our project, let me briefly introduce myself.

My name is Edina Suma. I am a licensed Pharmacy Technician, and I am currently pursuing my education as a Bachelor of Pharmacy student at the Faculty of Medicine, University of Prishtina.

My background in pharmacy has developed my interest in medicinal plants, active compounds, pharmacological effects and the relationship between chemical structures and biological activity.

My teammate comes from a different but equally important field: DevOps Engineering, with a background in software infrastructure, automation and system development.

Although pharmacy and technology may seem like two different worlds, we believe that combining them creates an opportunity to approach scientific challenges from different perspectives.

Now, let me start with one question:

What if we could use the genetic information of medicinal plants to explore new possibilities in natural compound production?

Nature has already developed an incredible diversity of bioactive compounds. Different medicinal plants produce different substances through highly specialized genetic and metabolic processes.

But what if we could compare these processes and investigate whether selected genetic characteristics could theoretically be combined?

That is the idea behind our project: **PhytoFusion AI.**

2. The scientific problem

Medicinal plants are important natural sources of bioactive compounds.

However, different plant species have different genes, enzymes and metabolic pathways, which influence the compounds they naturally produce.

Understanding these differences requires combining information from genetics, molecular biology, biochemistry and pharmacology.

Our objective is to bring these different scientific perspectives together through an AI-powered computational platform.

We want to investigate whether comparative genomics and artificial intelligence can identify genetic characteristics from different medicinal plants that could potentially be introduced into one host plant.

3. Why did we select these two plants?

For our scientific case study, we selected two medicinal plants belonging to the same genus:

Scutellaria baicalensis and *Scutellaria barbata*.

We selected them because they have different natural-product profiles and interesting pharmacological properties.

First, *Scutellaria baicalensis* is our proposed host plant.

It is particularly known for flavonoids such as baicalin, baicalein and wogonin.

These compounds have been extensively investigated for their anti-inflammatory and antioxidant properties, particularly through their influence on inflammatory mediators such as IL-6 and TNF-alpha.

Our second plant, *Scutellaria barbata*, is particularly interesting because of its clerodane diterpenoids.

Its compounds have been investigated for potential antitumor activity, especially their ability to influence cancer cell proliferation and apoptosis in preclinical studies.

This gives us an interesting pharmaceutical research perspective: one plant is well known for its flavonoid profile, while the other has attracted particular attention for its diterpenoid compounds.

Importantly, these activities are research findings, not proof that either plant is an established treatment for these conditions.

4. Understanding their genetic differences

To understand our approach, we first need to understand three biological concepts: DNA, genes and chromosomes.

DNA contains the genetic information of an organism.

Genes are functional segments of DNA that provide instructions for producing functional RNA or proteins.

Chromosomes are structures in which DNA is organized.

An interesting difference between our selected species is their chromosome number.

Scutellaria baicalensis has 18 chromosomes in its diploid cells, while *Scutellaria barbata* has 26.

Both species also have published chromosome-level genome assemblies.

Although they belong to the same genus, their genome structures are not identical. This is why we need comparative genomic analysis instead of assuming that their genes will automatically function in the same way.

We are not combining entire chromosomes. We are investigating selected genetic characteristics within the context of the host plant's genome.

5. Our genetic research hypothesis

Our research focuses on genes associated with the biosynthesis of natural compounds.

For example, PAL, CHS, 4CL and CHI are associated with different steps of flavonoid biosynthesis.

However, our main research interest is the KPS gene family, which is involved in clerodane diterpene biosynthesis.

A 2023 study by Qiu and colleagues identified and functionally characterized different KPS enzymes in these two Scutellaria species.

One particularly interesting finding was that SbarKPS1 in *S. barbata* produces an intermediate associated with clerodane diterpene biosynthesis, while the corresponding SbaiKPS1 in *S. baicalensis* showed very low gene expression in the examined tissues.

This gives us a specific scientific question:

Could introducing or modifying the expression of a selected donor genetic characteristic from *S. barbata* theoretically alter the early stages of diterpenoid biosynthesis in *S. baicalensis*?

Of course, introducing one gene does not automatically produce a complete bioactive compound.

The host plant would also need the necessary downstream enzymes, precursors and metabolic mechanisms.

Our goal is to investigate these requirements rather than assume that genetic integration will automatically produce a therapeutic effect.

6. Transition to Speaker 2

So, we are not simply interested in identifying genes.

We want to understand their functions, compare their biological roles and investigate their potential compatibility.

And this is where artificial intelligence and computational biotechnology become essential.

Let me hand over to my teammate, who will explain how we transform this scientific question into a technological solution.

PHYTOFUSION AI – SPEAKER 2

From Scientific Data to Computational Biotechnology

1. The connection between Pharmacy and DevOps

Thank you, Edina!

As mentioned, I come from the field of DevOps Engineering, where I work with software infrastructure, automation and system development.

In this project, my role is to help transform scientific information into a structured, reliable and accessible technological solution.

Pharmacy helps us understand the biological meaning of the data: the genes, active compounds, enzymes and their potential pharmacological relevance.

DevOps allows us to build the infrastructure that collects, processes, validates and organizes this information.

Through automation, data pipelines, testing and deployment, we can make complex scientific workflows more reproducible and accessible.

Pharmacy provides the scientific foundation, while DevOps helps turn that knowledge into a working technological solution.

2. How does PhytoFusion AI work?

Our platform is designed around five interconnected computational engines.

The first is **Scientific Data Ingestion**.

It collects genomic information, gene annotations and scientific publications from established biological databases.

The second is the **Comparative Genomics Engine**.

It compares genetic and protein sequences from our two selected plants, investigates gene relationships and identifies their chromosomal locations.

The third is **Pathway Intelligence**.

This engine connects genes to proteins, enzymes, biochemical reactions and natural compounds.

It helps us understand how individual genes contribute to larger metabolic pathways.

3. Our main innovation – GeneBridge AI

Our central innovation is GeneBridge AI.

We believe that comparing DNA sequences alone is not enough to determine whether two genetic characteristics are functionally compatible.

Two genes may have similar sequences but different biological functions.

Therefore, GeneBridge combines computational genomic analysis with scientific literature retrieval.

It investigates questions such as:

Are the selected genes genetically related?

Are their biological functions experimentally documented?

Does the host plant have the necessary metabolic pathway components?

Is there supporting gene expression evidence?

And what important information is still missing?

The system then generates an explainable scientific assessment supported by references.

We do not want an AI system that simply gives us an answer. We want an AI system that can explain why it reached that conclusion and show us the scientific evidence behind it.

4. CRISPR Scenario Simulator

Our fourth and fifth engines are GeneBridge AI and the CRISPR Scenario Simulator working together.

The simulator explores two hypothetical approaches.

The first is host gene modification, where we investigate how a selected modification to an existing host gene could potentially influence its metabolic pathway.

The second is donor genetic integration, where we evaluate the theoretical introduction of a selected genetic characteristic from the donor plant into the host genome using a CRISPR-assisted approach.

For example, we could investigate the potential integration of SbarKPS1 from *Scutellaria barbata* into *Scutellaria baicalensis*.

The simulator would assess its documented function, the host's existing genetic background, the relevant metabolic pathway and the potential biological limitations.

One important distinction is that CRISPR gene editing does not automatically mean successful integration of an entire donor gene.

Our platform does not perform actual genetic engineering. It provides a computational investigation of hypothetical scenarios.

Even if a genetic modification were technically successful, we would still need laboratory studies to confirm gene expression, identify the resulting compounds and investigate their pharmacological activity.

5. What will users actually see?

Through our web application, users will be able to explore six main sections.

A dashboard presenting both plant species and their scientific information.

A Genome Explorer displaying chromosome organization and selected gene locations.

GeneBridge for comparative genetic analysis.

A Pathway Explorer connecting genes, enzymes and natural compounds.

CRISPR Studio for exploring hypothetical genetic modifications.

And finally, a Virtual Plant Report presenting the proposed genetic scenario, supporting scientific evidence, predicted consequences and limitations.

This creates a complete research journey, from genomic information to an evidence-based engineering hypothesis.

6. Impact and closing

So, what makes PhytoFusion AI different?

We are bringing together three important fields: plant genetics, computational biotechnology and artificial intelligence, with a clear pharmaceutical research perspective.

Instead of looking at genes as isolated sequences, we investigate them as components of complex biological systems.

We also believe that scientific innovation must be responsible.

We will not claim that a virtual model represents a successfully engineered plant or that an altered natural-product profile automatically means improved therapeutic effectiveness.

Those questions require further experimental and pharmacological validation.

Our ambition is to create a platform that makes complex genetic information more accessible, understandable and useful for future research.

Because sometimes, the next scientific breakthrough does not begin with a ready-made answer.

It begins with the right question, the right scientific evidence and a new way of connecting them.

We are **PhytoFusion AI**.

Connecting plant genetics. Exploring new possibilities. Building the future of biotechnology.

Thank you!